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ORIGINAL RESEARCH

Day-ahead continuous double auction-based peer-to-peer energy trading platform incorporating trading losses and network utilisation fee

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Abstract

Integration of distributed energy resources, such as photovoltaic solar (PV), introduces new opportunities to establish local energy market frameworks to improve renewable energy utilisation in residential sectors. Such peer-to-peer (P2P) energy trading refers to a local market structure where customers (and prosumers) interact to share excess PV generation to enhance the individual and community social welfare. In this work, a day-ahead continuous double auction (CDA)-based P2P market structure considering network losses and network utilisation fees was designed. Day-ahead PV energy is modelled using fractional integral polynomials and the output is forecasted using an autoregressive integrated moving average model for each market interval. Based on the customer load and excess PV energy, the CDA market is cleared using a bid/ask matching mechanism. The performance of the P2P market was evaluated by computing different welfare metrics while analysing the effect of network constraints. The results show that the designed CDA-based P2P market structure increases the social welfare of all participants by an average of 17.75% compared to the baseline for the presented cases. Moreover, the impact of the forecasting error between the day-ahead and real-time market was also quantified.

KEYWORDS

energy management, microgrid, nanogrid, peer-to-peer energy trading, power markets, weather forecasting

1 | INTRODUCTION

The inflexibility of regulated monopolies for electric utilities with fixed retail/feed-in tariffs (FIT) for residential customers creates an opportunity for the formulation of a local market framework to enable a competitive market environment [1–3]. Residential customers that invest in rooftop photovoltaic solar (PV) can act as small-scale energy producers in the neighbourhood, enabling them to share excess energy with their neighbours. This gives rise to the concept of the *prosumer*—customers having renewable resources/battery energy storage systems (BESS) that can sell and consume energy at different market intervals [4, 5]. P2P energy trading refers to a market structure where the customers and prosumers interact to enhance the welfare of all market participants. In a traditional structure, prosumers sell their excess PV generation at the FIT, while customers purchase their demand at the utility price. This results in limited flexibility for both customers/prosumers due to the rigid nature of such tariffs (note that this also applies to time-of-use and real-time pricing). This reduces the social welfare of both the residential consumers/prosumers interacting with the electric utility. An alternative is to introduce an additional trading platform in the form of a P2P market where participants adjust their price signals according to their individual preferences [6, 7].

P2P markets can be designed either by using a centralised or decentralised approach [8, 9]. In a centralised structure, a central operator such as the distribution system operator

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(DSO) or a aggregator overlooks the transactions between the market participants [10, 11]. Contrarily, the decentralised market allows autonomous interactions between the customers and prosumers without requiring any central entity [12, 13]. An important factor in setting up the P2P trading platform is the design of the market framework and bi-lateral contracts between the customers and prosumers. A number of methods, such as blockchain [14], game theory [15], and optimisation-based methods [16] have been presented in the literature to develop P2P trading platforms. The selection of the method depends primarily on the nature of the objective function, bi-lateral contracts, and P2P market model.

The main challenge of setting up the P2P platform is to optimally decide the trading strategies between market participants to enhance the overall social welfare of the customers. Additionally, the inclusion of network constraints and the selection of forecasting algorithms to model the demand/supply curves of the participants can greatly influence the welfare of the P2P market. In this study, we provide a detailed design of a P2P trading platform while discussing the significance of network constraints and forecasting models for the performance metrics of the market.

1.1 | Literature review

This part of introduction presents a brief overview of the different types of the platforms introduced for P2P trading in literature. Additionally, the effect of network constraints and the forecasting methods is discussed in the subsequent subsections.

1.1.1 | Formulation of P2P trading platforms

Various market frameworks have been developed in the literature for P2P trading platforms based on different pricing strategies and network constraints. P2P trading has been discussed both in a standalone fashion (optimising the performance of a P2P trading market) and in combination with the other conventional optimisation problems such as demand response and BESS optimisation.

For example, in Ref. [17], the authors have presented a demand response-based P2P energy trading platform for customers and prosumers. The developed optimisation framework aims to minimise the electricity cost for all participants of the market while scheduling their flexible loads to off-peak periods. However, the authors have not considered in detail the effect of the network constraints and the trading strategies on the P2P market performance. Additionally, the clearing price for the participants is defined as a fixed percentage of the solar tariff, which may not be a realistic (or flexible) approach for actual markets. The authors in Ref. [18] introduced an intra-day model while incorporating the demand response mechanism for their P2P trading network. The intra-day trading platform was developed to allow the participants to account for any real-time mismatches to incorporate uncertainties in the renewable energy generation.

The authors in Ref. [19] suggested a locational marginal pricing-based P2P structure while considering a BESS. The framework simulates a day-ahead Optimal Power Flow (OPF) problem to determine the distribution marginal prices and the transaction fee for each prosumer in the network. Similarly, the authors in Ref. [20] have suggested the shared BESS for P2P trading platforms that uses a central storage unit for the prosumers to curtail the peak demand on the system. A bi-level leader-follower-based optimisation framework is presented in Ref. [21] for prosumers participating in the P2P market to optimise the performance of the P2P market. The suggested model solves the day-ahead market problem while considering the interactions of the prosumers with both the customers and the DSO. However, the authors have not considered in detail the bidding strategies and clearing price mechanism for the prosumers.

1.1.2 | Network constraints for P2P markets

An important aspect of P2P platforms is the impact of network constraints on market performance. Various techniques have been presented in the literature to model the physical constraints of the network for P2P markets. The authors in Ref. [22] have presented a P2P trading platform while considering the network constraints for the system. However, the suggested platform in Ref. [22] has not considered network constraint impacts for a CDA-based market. Additionally, the network constraints were not formulated for individual welfare metrics of the participants, rather a social economic welfare problem was presented.

The authors in Ref. [23] created a distribution locational marginal pricing-based transactive energy platform for prosumers to account for congestion management in distribution systems. The authors in Ref. [24] modelled a constrained CDA-based market for low voltage networks. However, transaction fee constraints have not been considered, which play a pivotal role in determining the welfare of market participants. A reserve constraint-based P2P platform was suggested in Ref. [25] to develop the interactions between prosumers and the DSO for finalising the market operations.

1.1.3 | CDA-based P2P markets

Another important consideration in designing P2P markets is the design of bidding strategies and transaction rules for the market participants. One of the most commonly used, yet efficient, way of generating and matching the P2P contracts is through an auction mechanism. To model the P2P market by using an auction based mechanism, the authors in Ref. [26] have suggested a Continuous Double Auction (CDA)-based P2P market framework. The authors compared different trading strategies for the CDA-based market while considering long-term P2P contracts. This research uses a similar CDA trading platform as suggested in Ref. [26] to set up the P2P energy transactions between the customers and the prosumers.

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In Ref. [27], a P2P energy trading platform to optimise the energy allocation for urban community microgrids was suggested for the trading mechanism. The authors presented a distributed energy management model, which preserves the privacy aspects of the market while using the CDA-based market for prosumers. Similarly, the authors in Ref. [28] have used a CDA-based P2P trading platform to simulate a case study in Japan while considering PV systems and plug-in hybrid electric vehicles. The framework facilitates the participation of prosumers/customers in a user-centric environment to enhance their welfare. A k-continuous-based CDA structure to simulate the P2P market for low-voltage distribution systems was presented in Ref. [29]. The major contribution was the unique clearing price mechanism for each participant of the market depending on the CDA market k-factor. The suggested CDA based market, however, again neglects the impact of network constraints on the trading mechanism.

For implementing the CDA-based market by using a game theoretic approach, the authors in Ref. [30] implemented Stackelberg's competition to determine the individual preferences for each customer/prosumer in the network. Additionally, the authors utilised a blockchain-based implementation for the CDA-based P2P framework. Different auction mechanisms such as one-sided, Vickrey, and generalised second price auctions are compared in Ref. [31] to set up a P2P market for different numbers of buyers and sellers. A dual blockchain implementation for a P2P trading platform was developed in Ref. [32] that consists of a primary blockchain for storing the P2P data and a secondary blockchain to perform the computationally expensive tasks, such as the pricing mechanisms and transaction execution.

1.1.4 | Dependence of P2P markets on load and PV models

Another important component of the P2P market is the impact and dependence of input parameters, such as demand and generation profiles of the market participants. The majority of the mentioned implementations in the above literature survey does not take into account the methodology required to forecast the demand/supply curves while considering individual participant behaviour. This can impair the performance of the day-ahead markets if the intermittent nature of the renewable energy resources is unaccounted for. Additionally, the load profile for each customer depends on the individual consumption pattern and seasonal variations. A general methodology that can capture the seasonal effect and the individual behaviour of the participants can improve the performance of day-ahead markets by a substantial factor.

1.2 | Research gap and contributions

In the literature, the majority of P2P trading platform implementations use an auction-based market for developing the bids and ask prices. However, the performance of such markets have rarely been evaluated considering the physical constraints of the system. If unaccounted for, the inclusion of network constraints, like trading losses and network utilisation fees, reduces the performance of the CDA-based P2P market and limits the welfare of the market participants. While the work in Ref. [24] suggests network constraints for CDA in the form of losses, it does not consider the utilisation fee model constraint and the effect of network parameters on the individual welfare of market participants. Additionally, the models for generating the demand/supply curves for the market participants while considering the seasonal variations and individual consumption patterns have not been discussed in detail for P2P markets in literature. Based on the mentioned research gap, following are the list of the contributions of this paper:

- designed a day-ahead constrained CDA-based P2P platform by using a utilisation fee and quadratic trading loss model to evaluate both the individual and social welfare metrics;
- suggested a contribution factor-based electrical distance formulation for the utilisation fee model to individually elaborate the effect of the transaction fee on the social welfare of market participants;
- implemented the queueing load model to generate unique load curves for each customer/prosumer in the network to capture the effect of seasonal variations and behaviour of individuals; and
- developed a detailed PV forecasting model using fractional integral polynomials with an autoregressive integrated moving average (ARIMA) model for determining the day-ahead PV profile for each prosumer in the network.

The rest of the paper is organised as follows: Section 2 describes the system model. Section 3 presents the modelling of the P2P components. Section 4 presents the CDA-based P2P trading platform while incorporating the network constraints. Section 5 gives the brief overview of simulation parameters and the test case. Section 6 presents the results and discussion for different cases. Section 7 presents the conclusions of this work.

2 | SYSTEM MODEL

The proposed trading system consists of three main stages [26]. The first stage involves the detailed modelling of the P2P components. For developing the P2P market structure, a PV profile for each prosumer and load curves for all market participants are required. The combination of fractional integral polynomial and ARIMA models are used to determine the day-ahead forecast and real-time PV output power of each prosumer for different market intervals. For load profiling, the queueing load model is used.

The second stage establishes the P2P market by using the auction-based market while considering network constraints. Bids/ask prices are generated for each market participant, and orders are matched based on the Equilibrium Matching (EM) principle while considering trading losses and a network

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utilisation fee. The bid value refers to the price offered by the customer to purchase the energy, whereas the ask price refers to the value set up by the prosumer to sell the excess renewable power. The network consists of c customers and p prosumers that interact during each market interval. In all our discussion, the customers refer to the market participants who will consume the excess energy offered by the prosumers. The total market duration is divided into t trading intervals of equal duration.

The third stage implements a clearing mechanism for any unmatched orders for the CDA-based market. At the end of each trading period, any unmatched orders will be cleared by the DSO. The remaining load of consumers is satisfied using the retail price by the DSO. Contrarily, the DSO offers a FIT to prosumers to purchase the remaining PV generation. It should be noted that prosumers use PV energy first for self-consumption (more profitable as the retail price is always greater than FIT) and then trade with neighbours via the P2P market. The details of each stage are given in the subsequent sections. Figure 1 gives the breakdown of different stages of the proposed P2P trading platform.

3 | P2P COMPONENT MODELS

As shown in Figure 1, the first stage in the design of the P2P market is to model the PV profile for each prosumer and load curves for all the market participants. The queueing load model is used to capture the seasonal variations and human behaviour in electric demand, and fractional integral polynomials are implemented to model PV power based on temperature and solar irradiance. To forecast day-ahead irradiance and temperature values for the PV model, ARIMA is used. It is important to note that the designed P2P market framework will work with any choice of load, PV, and forecast models. The rest of this section provides details of each P2P market component model.

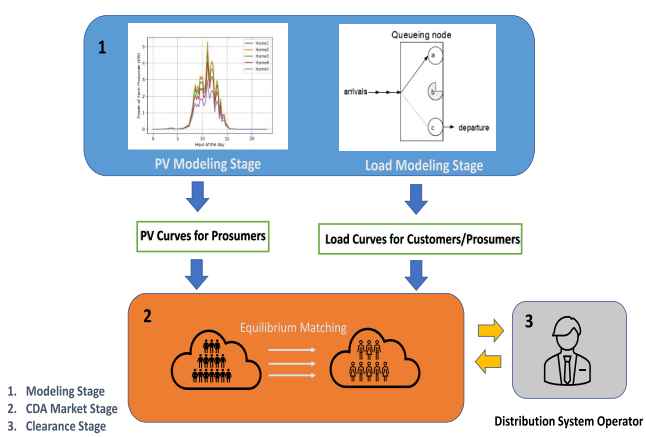

FIGURE 1 Breakdown of different stages of continuous double auction (CDA)-based market. The first stage models the load and photovoltaic (PV) curves for each market participant using the queueing load model and autoregressive integrated moving average (ARIMA) forecasting. In the second stage, the trading between the prosumers and customers is conducted based on the CDA market. The final stage clears unmatched orders with the distribution system operator (DSO).

3.1 | Queueing load model

The queueing load model is used to model the electric demand curves for the market participants [33, 34], which uses queueing theory initially created for the communication sector. The major components of a queueing model are the probability distributions defined for the service time and inter-arrival time, the queue capacity, the size of the population, and the number of servers. These characteristics can be defined as either time varying quantities or static in nature. Queueing theory is then used to model the household consumption profiles accordingly:

- The start time for each electric load is generated randomly according to the queueing model. Due to the fact that each residential customer uses appliances according to their own schedule, the random nature of the start time of the appliances has the desired characteristic that the individual behaviour of each market participant is unique. However, the queueing load model also guarantees the aggregated demand of the customers approaches a known load curve.
- A Poisson distribution is used to model the arrival rate of the appliances (e.g. start times), which captures the time-varying nature of the loads (daily and seasonal variations).
- Each appliance modelled using the queueing theory has distinct characteristics. For example, the modelled devices can be electrical, thermal, and/or schedulable.

The queueing model for developing the load profiles of the residential customers considers the following assumptions: (i) The number of the appliances that can arrive in the queue are infinite (size of the population is infinite). This assumption leads to the fact that the incoming appliances will not affect the number of the appliances already present in the queue; (ii) The length of the queue is infinite so there are no loss of appliances (one can think of this as no limit on the number of running appliances); and (iii) The service policy is first-come-first-served. The queueing model used in this research is developed in accordance with Ref. [34] and uses the characteristics as defined by $M_t/G/\infty$, where M_t represents the time varying Poisson process to model the inter-arrival time of the devices, G represents that the probability distribution used for the service time is considered to be general (electric load run times can follow any probability distribution or mass), and ∞ shows that the capacity for the individual customers is taken to be infinite. The time-varying Poisson process $\lambda(t)$ to model the appliance rate of the system is given as follows:

$$\lambda(t) = \frac{L(t + E[d])}{E[d]E[p]} \tag{1}$$

The individual household demand $L(t)$ is generated using the distribution load data $C_l(t)$. To scale $C_l(t)$ to the individual demand of the customers, the following relation is used:

$$L(t) = \beta_{\min} + \frac{C_l(t) - \min(C_l)}{\max(C_l) - \min(C_l)}.(\beta_{\max} - \beta_{\min}) \tag{2}$$

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The readers are encouraged to go through the mathematical details of the developed queueing load model presented in Ref. [34].

3.2 | PV model and day-ahead forecasting

In addition to the customer/prosumer load profiles, a PV model is required with a day-ahead forecast for each prosumer to run the P2P market. The output of a PV system is intermittent depending on external atmospheric conditions, that is, solar irradiance (W/m^2) and temperature (°C). A number of forecasting techniques and mathematical models have been presented in Refs. [35, 36] to forecast the renewable generation for different intervals. Accordingly, the following steps are used for modelling/determining PV power [37]:

1. Develop the PV model for each prosumer, in this work by using the fractional integral polynomial mathematical model;
2. Design an ARIMA model to forecast the day-ahead solar irradiance and temperature for each time interval of the P2P market; and
3. Compute the day-ahead PV generation based on the forecast atmospheric parameters, PV model, and rated system capacity for use in the P2P market.

3.2.1 | Mathematical PV system model

Each prosumer in the P2P market is assumed to own a PV system; this subsection describes that the mathematical model is used to determine the output of each PV system. We adopt the fractional and integral polynomial based method [38] to determine the PV power based on the solar irradiance and temperature. This proposed method has the following advantages:

- Use of both the irradiance and temperature curves to compute the PV power, where most literature assume standard test conditions (STC) of 25°C. The inclusion of the temperature parameter reduces the modelling error compared to actual PV systems.
- Accounts for other PV module characteristics, such as the open circuit voltage, short circuit current, the voltage coefficient, and temperature coefficient. This gives a practical approach to model a P2P community, as each prosumer in the network has different PV modules (vendor etc.) installed with distinct characteristics.

The suggested model finds the current i of the module as the function of the voltage v which is given by the following relation:

$$i(v) = i'_{sc} - i'_{sc}\left(\frac{v}{v'_{oc}}\right)^{a+b}, \tag{3}$$

where i'_{sc} and v'_{oc} represent the short circuit current and the open circuit voltage of the module at given irradiance and temperature levels, respectively. The value i represents the module current in the range of $0 \le i \le i'_{sc}$, and v represents the module voltage given in the range of $0 \le v \le v'_{oc}$. The value b is given in the range of $0 \le b < 1$, and a represents a non-negative integer. The power P of the module by using the above characteristics can be defined as follows:

$$P(v) = i(v) \cdot v = \left(i'_{sc} - i'_{sc}\left(\frac{v}{v'_{oc}}\right)^{a+b}\right)v \tag{4}$$

The characteristics as defined in Equation (3) depend on the three contributing parameters, i'_{sc}, v'_{oc}, and the sum $a + b$. These parameters can be determined using the following relations:

$$v'_{oc} = s_m V_t\left(\frac{G}{G_{stc}}\right)(T - T_{stc}) + v_{\max} - (v_{\max} - v_{\min}) \cdot e^{\left(\frac{G}{G_{stc}}.\ln\left(\frac{v_{\max}-v_{oc}}{v_{\max}-v_{\min}}\right)\right)} \tag{5}$$

$$i'_{sc} = p_m\left(\frac{G}{G_{stc}}\right)(i_{sc} + I_t(T - T_{stc})) \tag{6}$$

$$a + b = \frac{i_{sc}}{i_{sc} - i_{opt}}, \tag{7}$$

where s_m and p_m represent the number of series and parallel connected modules, respectively, G_{stc} and T_{stc} represent the irradiance and temperature levels at STC, respectively, and G and T are the given irradiance and temperature level, respectively. Additionally, v_{max} and v_{min} are the maximum and minimum voltage levels for the PV module, i_{opt} represents the optimal current value for the PV module, V_t represents the temperature coefficient value for the open circuit voltage, I_t represents the temperature coefficient value for short circuit current of the module, and v_{oc} and i_{sc} are the rated parameters of the module at STC, respectively.

3.2.2 | Day-ahead forecasting using ARIMA

To obtain the day-ahead temperature and solar irradiance to be used for the prosumer P2P market ask prices, a forecasting technique is required. The method chosen from literature to forecast the input parameters of the PV system is the ARIMA model [39] due to its ability to deal with the non-stationary data sets and seasonal variations. The three major components for the ARIMA(p,d,q) model are as follows: Autoregressive (AR) part defined by the order p, Moving Average (MA) model defined by the order q, and the order of differentiation d. The mathematical relations for the AR, MA, and ARMA models are given as follows:

$$x_t = \alpha + \sum_{k=1}^{p} \phi_k x_{t-k} + \epsilon_t \tag{8}$$

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$$x_t = \epsilon_t + \sum_{k=1}^{q} \delta_k \epsilon_{t-k} \tag{9}$$

$$x_t = \alpha + \epsilon_t + \sum_{k=1}^{q} \delta_k \epsilon_{t-k} + \sum_{k=1}^{p} \phi_k x_{t-k}, \tag{10}$$

where x_t represents the time series data, k represents a specific time series index, ϕ represents the parameters of the AR model, δ represents the parameters for the MA model, α represents a constant, and ϵ represents the white noise error terms. The above relations can be modified in terms of the lag operator $l^k(x_t) = x_{t-k}$ as follows:

$$\epsilon_t = \left(1 - \sum_{k=1}^{p} \phi_k l^k\right) x_t - \alpha \tag{11}$$

$$x_t = \left(1 + \sum_{k=1}^{q} \delta_k l^k\right) \epsilon_t \tag{12}$$

$$\left(1 - \sum_{k=1}^{p} \phi_k l^k\right) x_t = \left(1 + \sum_{k=1}^{q} \delta_k l^k\right) \epsilon_t + \alpha \tag{13}$$

The term d used in the ARIMA model is to deal with the non-stationary nature of the dataset. The complete ARIMA(p, d,q) model involving the differential term d is given as follows:

$$\left(1 - \sum_{k=1}^{p} \phi_k l^k\right)\left(1 - l^d\right) x_t = \left(1 + \sum_{k=1}^{q} \delta_k l^k\right) \epsilon_t \tag{14}$$

The unknown parameters in building the ARIMA model are p, d, and q. To optimally tune these parameters, the Box–Jenkins methodology is used [37]. For any given dataset, the following steps are used in the Box–Jenkins methodology: (i) Identification of the model; (ii) estimation of the parameters; and (iii) residual diagnostics. In the identification of the model, different plots like auto correlation function (ACF) and partial ACF are used to determine the nature of the dataset. To estimate the parameters, different statistical measures, such as Augmented Dickey–Fuller statistics and information criteria, such as Akaike and Bayesian criteria, can be used to determine the values of p, d, and q. For residual diagnostics, different plots such as the density plot for the residual terms and ACF plots can be utilised to validate the model. For the sake of brevity, the details of these steps are not elaborated here, but readers are encouraged to go through the details of each of these steps found in Refs. [39, 40].

3.2.3 | PV system power computation

Based on the day-ahead forecast climate parameters and the developed PV model, the power for each P2P trading interval is computed using the following assumptions:

- Forecast irradiance and temperature levels remain constant for each interval t and will update at the start of next interval $t + 1$.
- For a given irradiance and temperature level, the power of the module is determined using the developed fractional and integral polynomial model given by Equations (3)–(7).
- Maximum power point operating condition is considered for each PV module.
- The total number of the modules required can be computed based on the rated system capacity and the rated maximum power of the individual module.
- The total efficiency for the converter system is considered to be 95% [37].

This concludes the forecasting steps required to determine the PV power for different market intervals and the modelling of the P2P components. The next section describes in detail the steps required to setup the market under the network constraints.

4 | CDA-BASED MARKET FRAMEWORK FOR P2P TRADING

Once the electric demand and PV energy per time interval are calculated for each participant, the next in the P2P framework is to set up the market based on CDA auctions. CDA is a platform where multiple sellers and buyers participate and compete to trade a certain commodity. In our case, the buyers are customers willing to purchase excess PV generation, whereas the sellers are prosumers that want to sell their excess PV power. The main components of the CDA-based P2P market are the generation of the customer bids and prosumer ask values, the matching mechanism, and the clearing strategy. This section provides complete details regarding the modelling of each component and incorporates the effect of network constraints on the traditional CDA market, with pseudocode of the market process provided in Algorithm 1.

4.1 | Generation of bid/ask signals

The first step of the CDA market is to generate the bid and ask prices for each participant by using a certain trading strategy. A number of methods such as the Zero Intelligent (ZI) strategy, Eyes on Best Price strategy, and ZI Plus trading mechanism [41] have been suggested in the literature to model the market price signals. This research uses the ZI strategy to initialise the bid/ask prices for each market participant, but note that the designed framework will work with any bidding strategy. The ZI strategy considers the fact that each market participant does not consider prior market experience and knowledge to generate their bid/ask values, leading to the generation of the bids using a random criterion. In our proposed design, the generation of the bid/ask values for market participants is modelled using a uniform random number within the range of the utility price and FIT. Mathematically, the market price

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signals using the ZI strategy for any customer or prosumer can be given as follows:

$$B_{ask,\Delta t} = \mathcal{U}\left(T_{\text{feed-in},\Delta t}, T_{retail,\Delta t}\right) \tag{15}$$

$$B_{bid,\Delta t} = \mathcal{U}\left(T_{\text{feed-in},\Delta t}, T_{retail,\Delta t}\right), \tag{16}$$

where at time interval Δt, $B_{ask,\Delta t}$ and $B_{bid,\Delta t}$ represent the ask and bid values, respectively, and $T_{feed-in,\Delta t}$ and $T_{retail,\Delta t}$ represent the FIT and retail price, respectively. One of the main advantages of the proposed ZI strategy is that each P2P participant will get a competitive price compared to the DSO market. This is due to the fact that the random initialisation of the bid/ask values within the range of retail and FIT ensures a positive welfare for both the customers and prosumers.

Figure 2 describes the mechanism for the generation of the bids/ask values using the ZI strategy. Due to the constrained nature of the proposed ZI strategy between the retail and FIT, the customer cannot place a bid value that is lower than the FIT. Similarly, the prosumer cannot initialise ask prices higher than the retail value for any market interval. This mechanism ensures that for each matched P2P contract, the customer will receive a local market price competitive to the retail price for the given market interval. Similarly, the prosumer will sell its surplus PV generation at a price that is higher than the FIT of the market. This results in a positive social welfare for all the market participants by ensuring certain monetary advantages. The complete mathematical details for the matching of orders and computing the P2P price are given in the subsequent subsections.

Algorithm 1 CDA-based P2P energy trading pseudo code

Data: Retail price, FIT, demand and PV curves
Result: Welfare Metrics
$\Delta t \leftarrow 1$
while $\Delta t \leq t$ **do**
  Compute $\tau_{t,(c\times p_r),\Delta t}$
  for $i \leftarrow 1$ **to** c **do**
    for $j \leftarrow 1$ **to** p_r **do**
      Determine $G_{j,\Delta t}$
      Match orders based on equilibrium matching
      if $b_{i,\Delta t} \geq a_{j,\Delta t}$ **then**
        $c_{p,ij} = \frac{1}{2}(b_{i,\Delta t} + a_{j,\Delta t})$
        $p_{ij} = \min(D_{i,\Delta t}, G_{j,\Delta t})$
      if $b_{i,\Delta t} < a_{j,\Delta t}$ **then**
        $c_{p,ij} = 0$
        $p_{ij} = 0$
  Determine clearing quantities
  Determine C_c and C_p
  Determine $C_c^{'}$ and $C_p^{'}$
  Determine η_c, η_p, and η_{comm}
  $\Delta t \leftarrow \Delta t + 1$

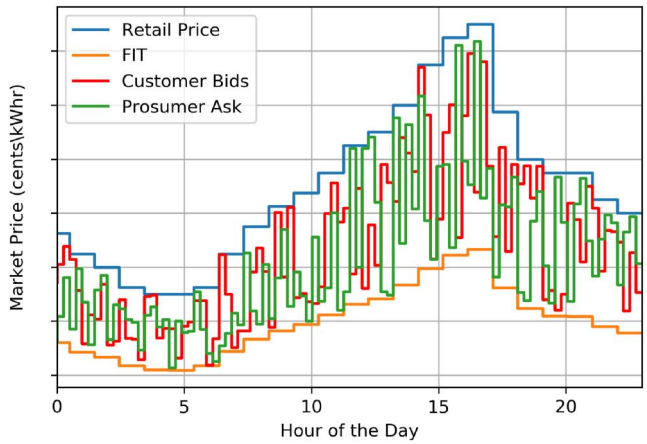

FIGURE 2 Initialisation of the bids/ask values using zero intelligent strategy. FIT, feed-in tariff.

4.2 | Matching and clearing mechanisms

After generating the price signals for each customer/prosumer in the network, the next step in designing the CDA-based P2P market is to define the matching mechanism. This research uses the Equilibrium Matching (EM) to match the P2P [42, 43]. In EM, the bid values are sorted in descending order, and the ask prices are sorted in ascending order. The matching between the customer and the prosumer occurs when the bid price is greater than or equal to the ask price of the prosumer (lines 8–10 of Algorithm 1). This process continues and all the matched orders will progressively be cleared till there are no further matched orders (lines 4–13 of Algorithm 1). At the end of each market interval, all the unmatched orders will be cleared by the DSO. Each prosumer will be offered the FIT to clear any remaining PV generation, and each customer will be offered the utility price to settle any remaining demand value.

The complete mathematical details of the matching and clearing mechanism are as follows: Let us consider a network of $P = \{1, 2, 3, \ldots, p_r\}$ prosumers and $C = \{1, 2, 3, \ldots, c\}$ customers participating in the P2P market. Let $T = \{1, 2, 3, \ldots, t\}$ be the market intervals where trading between the participants will occur. For any given trading interval Δt, the trading matrix τ_t for p_r prosumers and c customers is given as

$$\tau_{t,(c\times p_r),\Delta t} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & \cdots & p_{1p_r} \\ p_{21} & p_{22} & p_{23} & \cdots & p_{2p_r} \\ p_{31} & p_{32} & p_{33} & \cdots & p_{3p_r} \\ p_{41} & p_{42} & p_{43} & \cdots & p_{4p_r} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ p_{c1} & p_{c2} & p_{c3} & \cdots & p_{cp_r} \end{bmatrix} \tag{17}$$

In the above matrix, the trading quantity $p_{ij} \in \tau_t$ for a matched order between the ith customer and the jth prosumer is given as follows:

$$p_{ij} = \begin{cases} \min\left(D_{i,\Delta t}, G_{j,\Delta t}\right) & \text{if } b_{i,\Delta t} \geq a_{j,\Delta t} \\ 0, & \text{if } b_{i,\Delta t} < a_{j,\Delta t}, \end{cases} \tag{18}$$

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where $D_{i,\Delta t} \forall i \in C$ and $G_{j,\Delta t} \forall j \in P$ represent the demand of the ith customer and excess PV generation of the jth prosumer, respectively, for a given trading interval Δt. Additionally, for the given trading interval Δt, $b_{i,\Delta t}$ and $a_{j,\Delta t}$ represent the bid value for the ith customer and ask price for jth prosumer, respectively, and the excess PV generation $G_{j,\Delta t}$ can be calculated as

$$G_{j,\Delta t} = \begin{cases} P_{j,\Delta t} - D_{j,\Delta t} & \text{if } P_{j,\Delta t} > D_{j,\Delta t} \\ 0 & \text{otherwise.} \end{cases} \tag{19}$$

Here, for the trading interval Δt, $P_{j,\Delta t}$ represents the forecast power and $D_{j,\Delta t}$ represents the demand of the jth prosumer, respectively. The trading price $c_{p,ij}$ for a matched order between the ith customer and jth prosumer is given as

$$c_{p,ij} = \begin{cases} \frac{1}{2}\left(b_{i,\Delta t} + a_{j,\Delta t}\right) & \text{if } b_{i,\Delta t} \geq a_{j,\Delta t} \\ 0, & \text{if } b_{i,\Delta t} < a_{j,\Delta t}. \end{cases} \tag{20}$$

For a given matched order, the trading price $c_{p,ij}$ for a given interval Δt is calculated as the average of the submitted bid $b_{i,\Delta t}$ by the customer and submitted ask price $a_{j,\Delta t}$ by the prosumer. For any interval Δt, the total quantity traded by the ith customer and jth prosumer in the network is given as follows:

$$E_{i,T} = \sum_{n=1}^{p_r} p_{in}, \ \forall \ i \in C \tag{21}$$

$$E_{j,T} = \sum_{k=1}^{c} p_{kj}, \ \forall \ j \in P, \tag{22}$$

where $E_{i,T}$ and $E_{j,T}$ represent the total quantity traded by the ith customer and jth prosumer, respectively. For the clearing mechanism at the end of the given market interval Δt, the DSO clears any unmatched orders by using the utility price and FIT. The clearing quantities can be determined using the following relations.

$$E_{i,cl} = D_{i,\Delta t} - \sum_{n=1}^{p_r} p_{in}, \ \forall \ i \in C \tag{23}$$

$$E_{j,cl} = G_{j,\Delta t} - \sum_{k=1}^{c} p_{kj}, \ \forall \ j \in P, \tag{24}$$

where $E_{i,cl}$ and $E_{j,cl}$ represent the clearing quantities for the ith customer and jth prosumer, respectively, for the market interval Δt.

4.3 | Network constraints

The ideal CDA market suggested in the literature does not consider the effect of the network constraints on the welfare metrics. However, in the practical networks certain limiting factors such as the voltage constraints, thermal line limits, network losses, and network utilisation fees can impair the performance or render local markets technically infeasible. This research considers two physical network factors for P2P trading, the trading losses considered for each transaction (i.e. line losses based on electrical distance between the buyer and seller), and the network utilisation fee paid to DSO for each matched order (i.e. to recoup capital costs of the physical network infrastructure).

4.3.1 | CDA trading losses

In actual test systems, the P2P energy trading takes place over the physical distribution network. This results in a certain amount of energy loss for each transaction between the customer and the prosumer. This research uses the quadratic loss function which models the P2P network losses as the quadratic function [22] of the quantity p_{ij} traded between the ith customer and jth prosumer and is given as follows:

$$P_L = \Gamma_{ij} p_{ij}^2, \tag{25}$$

where Γ_{ij} represents the loss coefficient for the given network. The value of the Γ_{ij} can be determined based on the network configuration (e.g. impedance network). For each matched order, a certain amount of trading loss would be included to the total amount of the traded quantity p_{ij}. Although in this first work we are working on a small neighbourhood network, this trading loss can be extended for any size distribution network.

4.3.2 | Network utilisation fee

In the P2P framework, the trading between customers and prosumers takes place over the lines owned by the DSO, therefore the participants need to pay a certain amount for market services provided by the DSO. This gives rise to the concept of a network utilisation fee, which is the amount paid by the market participants to the DSO for its market services and to recover capital expenditure. This research proposes an impedance-based electrical distance computation for developing the utilisation fee model for the P2P market. The reason for selecting the impedance-based method is due to its ability to compute the electrical distance by using the network topology for the distribution/radial systems. The suggested method ensures that transactions taking place over longer distances (higher losses through DSO owned infrastructure) results in a higher utilisation fee for market participants [44]. The electrical distance between the ith customer and the jth prosumer in a network is given by the modulus of the equivalent impedance as follows:

$$d_{ij} = |Z_{ij,equ}| = |z_{ii} + z_{jj} - 2z_{ij}|, \tag{26}$$

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where d_{ij} is the electrical distance between the ith customer and the jth prosumer, $Z_{ij,equ}$ is the equivalent impedance, z_{ii} and z_{jj} represent the self-impedance of the ith customer and the jth prosumer in the node impedance matrix, and z_{ij} is the mutual impedance in the node impedance matrix. Based on the computed electrical distance, the network utilisation fee paid to the DSO for each P2P transaction is given by the following equation:

$$F(p_{ij}) = \underbrace{\alpha_c(\gamma_i d_{ij} p_{ij})}_{\text{customers fee}} + \underbrace{(1-\alpha_c)\left(\gamma_j d_{ji} p_{ij}\right)}_{\text{prosumers fee}} \tag{27}$$

where α_c given in the range $0 \le \alpha_c \le 1$ gives the contribution factor for each transaction paid by the customer/prosumer to the DSO. The value of α_c can be pre-negotiated in the form of the P2P contracts with the market participants or it can be determined based on the network configuration. However, for the sake of simplicity, we will use the constant value of α_c for all P2P transactions. Let γ_i and γ_j represent the charge rate coefficients for the ith customer and jth prosumer, respectively. The overall fee paid to the DSO is then given as

$$TF(p_{ij}) = \underbrace{\sum_{i\in C}\left(\sum_{j\in P}\alpha_c(\gamma_i d_{ij} p_{ij})\right)}_{\text{network fee for customers}} + \underbrace{\sum_{j\in P}\left(\sum_{i\in C}(1-\alpha_c)\left(\gamma_j d_{ji} p_{ij}\right)\right)}_{\text{network fee for prosumers}}, \tag{28}$$

where $TF(p_{ij})$ represent the total network utilisation fee paid to the DSO for each trading interval Δt.

4.4 | Welfare metrics

The performance of the CDA-based market is evaluated using social welfare metrics. The following three welfare evaluators are used to compare the performance of the CDA-based market with the traditional market: (i) *customer welfare* that computes the decrease in payments of customers compared to the traditional market; (ii) *prosumer welfare* that computes the increase in profit of prosumers compared to the traditional market; and the (iii) *community welfare* that determines the overall welfare of the society by taking into account the market price for both the customers and prosumers. The main objective of the designed P2P market is to enhance these metrics, with a higher welfare equating to a better designed market mechanism. Mathematically, the following relations are used to compute each welfare metric.

$$C_c = \sum_{i\in C} D_{i,\Delta t} T_{\text{retail},\Delta t} \tag{29}$$

$$C_p = \sum_{j\in P} G_{j,\Delta t} T_{\text{feed-in},\Delta t}, \tag{30}$$

where C_c and C_p are the original market prices for customers and prosumers, respectively, without P2P trading. The updated market prices after incorporating the P2P trading are given as follows:

$$C'_{\text{c}} = \sum_{i\in C}\left(\sum_{j\in P} c_{p,ij} p_{ij} + E_{i,cl} T_{\text{retail},\Delta t}\right) \tag{31}$$

$$C'_{\text{p}} = \sum_{j\in P}\left(\sum_{i\in C} c_{p,ij} p_{ij} + E_{j,cl} T_{\text{feed-in},\Delta t}\right) \tag{32}$$

Here, C'_{c} and C'_{p} represent the market prices with P2P trading for customers and prosumers, respectively. Based on the computed network fee utilisation model, the updated P2P prices are as follows:

$$C'_{\text{c}} = \sum_{i\in C}\left(\sum_{j\in P} c_{p,ij} p_{ij} + E_{i,cl} T_{\text{retail},\Delta t}\right) + \underbrace{\sum_{i\in C}\left(\sum_{j\in P}\alpha_c(\gamma_i d_{ij} p_{ij})\right)}_{\text{network fee for customers}} \tag{33}$$

$$C'_{\text{p}} = \sum_{j\in P}\left(\sum_{i\in C} c_{p,ij} p_{ij} + E_{j,cl} T_{\text{feed-in},\Delta t}\right) - \underbrace{\sum_{j\in P}\left(\sum_{i\in C}(1-\alpha_c)\left(\gamma_j d_{ji} p_{ij}\right)\right)}_{\text{network fee for prosumers}} \tag{34}$$

The welfare metrics using the above relations can finally be computed as

$$\eta_c = \frac{C_c - C'_{\text{c}}}{C_c} \times 100\% \tag{35}$$

$$\eta_p = \frac{C'_{\text{p}} - C_p}{C_p} \times 100\% \tag{36}$$

$$\eta_{comm} = \left|\frac{(C_c - C_p) - \left(C'_{\text{c}} - C'_{\text{p}}\right)}{(C_c - C_p)}\right| \times 100\%, \tag{37}$$

where η_c, η_p, and η_{comm} represent the customer, prosumer, and community welfare, respectively.

5 | SIMULATION TEST CASE

The developed P2P market framework is simulated on a distribution network as shown in Figure 3. The developed network consists of $p_r = 10$ prosumers and $c = 5$ customers placed randomly at different nodes in the network. The complete

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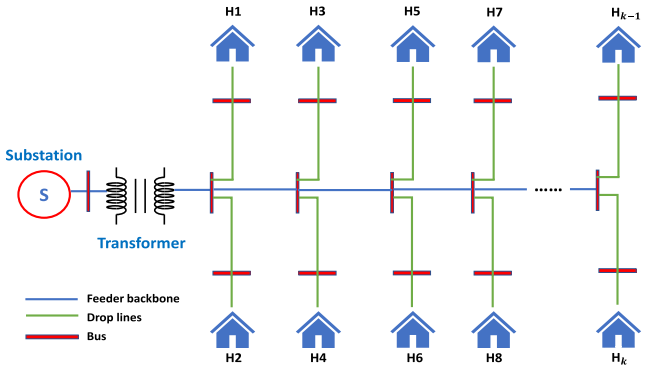

FIGURE 3 Simulation network for peer-to-peer trading incorporating the network constraints. The network is developed in accordance with the parameters given in Ref. [45].

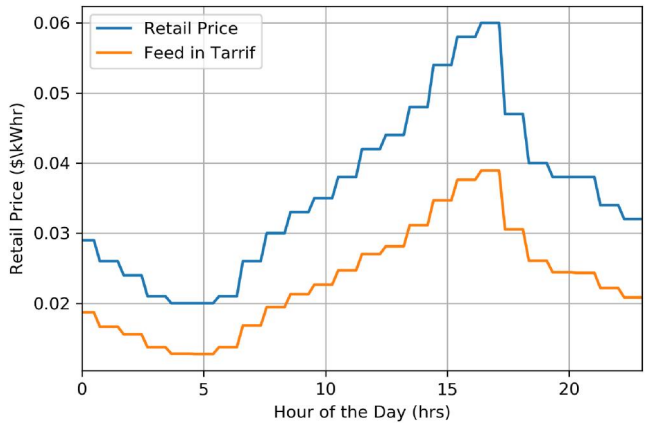

FIGURE 4 Retail and feed-in tariff offered by the distribution system operator to the market participants.

details regarding the network parameters of the given feeder are provided in Ref. [45]. The total trading period is simulated over a single day having $T = 96$ intervals of equal duration (15-min). The following subsections provide the details regarding the data used for each stage.

5.1 | P2P modelling data

The climate dataset used for the PV system models was taken from the National Renewable Energy Laboratory [46] for two consecutive years (January 2014–December 2015) at Los Angeles, California to forecast the irradiance and temperature values. The first 75% of the data was used to train the ARIMA model, while the remaining 25% was used for testing and validation. For the simulation, a day-ahead forecast was obtained for 1 January 2016. To model the PV systems for each prosumer, the module characteristics were used from Ref. [38]. Additionally, the rated system capacity for each prosumer was randomly initialised in the range [3, 6 kW] [26]. For the queueing load model, the distribution system aggregate data was taken from the Pennsylvania, New Jersey, and Maryland independent system operator given in Ref. [47] for 1 January 2016 for the ComEd load area. The values used for β_{min} and β_{max} in case of the queueing model are 500 W and 5 kW, respectively. The total number of the appliances considered are 15. For each device, the mean power is $\mu_p = 1$ kW and the standard deviation is $\sigma_p = 250$ W. The values for the queueing load model were taken from Ref. [34].

5.2 | Market parameters

The total market duration was taken to be 24 h divided into 96 equal intervals of equal duration. The FIT and retail tariff offered by the DSO to the market participants are taken to be time-varying and shown in Figure 4. The tariff is taken from the ComEd price data [48]. The network loss coefficient was taken to be 5% of the trading losses. The market contribution factor α was taken to be 0 (the prosumers will be charged for the P2P transactions in this study). The charge rate coefficient for each customer and prosumer was taken to be 0.03 for each P2P transaction ($\gamma_i = \gamma_j = 0.03$) [22]. Because the ZI strategy initialises the bid/ask prices randomly, to introduce fairness to all market participants, the results of the P2P market are averaged over N number of samples. The total number of samples taken for simulating each trading interval were $N = 1000$.

In addition to the 24-h market, each test case developed for the P2P market was compared in depth for two intervals. The first interval simulated the P2P market where the excess PV generation of the prosumers was relatively lower than the customer demand. The second interval was simulated where the excess PV generation was relatively higher than the demand. All the simulations are performed on Python using the Intel(R) Core(TM) i7-7500U CPU with a 2.70 GHz processor and 8.00 GB installed RAM.

Finally, for the two intervals, comparisons were made for each of the following cases:

- Case I: P2P trading without network constraints
- Case II: P2P trading with trading losses
- Case III: P2P trading with the network utilisation fee
- Case IV: P2P trading with trading losses and the network fee.

6 | RESULTS AND ANALYSIS

This section provides the detailed results and analysis of the different stages of the suggested P2P framework. First, the results for the modelling part of the P2P components are presented. Then, the detailed results for the P2P market using different test cases will be discussed.

6.1 | PV forecasting and queueing model

The first part of the modelling stage is to determine the load profiles for each customer/prosumer in the network by using the queueing load model. The queueing load model was

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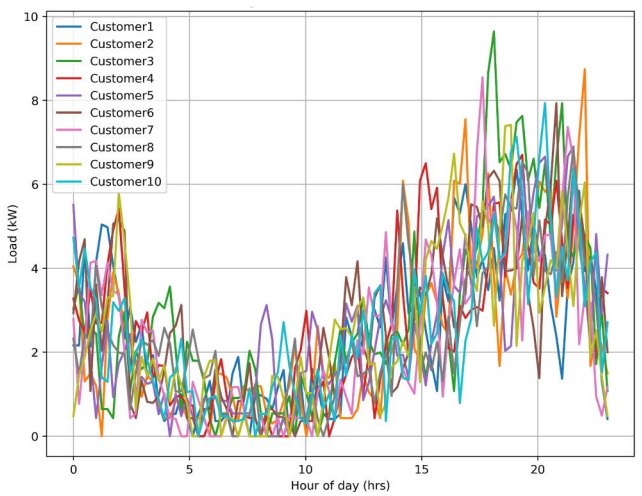
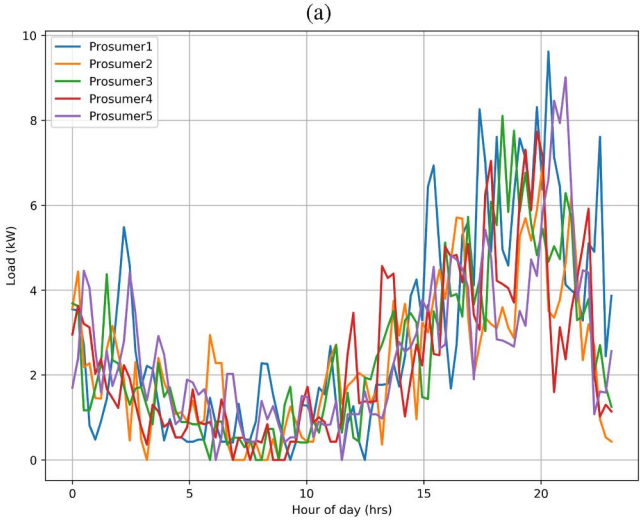
implemented using the parameters provided in Section 5, and the results are shown in Figure 5. The maximum capacity for the input load to the queueing model defined for each prosumer and customer in the network was 5 kW, however, for certain trading intervals, the queueing model exceeds the rated capacity due to the 'burstiness' effect of the $M_t/G/\infty$ model. Because of the randomness of the Poisson process inter-arrival times, a large number of appliances can arrive during a very short time interval that may result in unrealistic peaks. If necessary, an implementer of this P2P framework could limit this overshoot via the $M_t/G/C$ model, where C defines a hard limit on defined capacity for each household in the network [34].

To determine the forecast irradiance and temperature parameters for computing the PV power for each trading interval, the Box–Jenkins methodology was implemented using the dataset mentioned in Section 5. The final model selected based on the methodology defined in Section 3 was ARIMA(4,1,5)—four AR terms, one differentiation term, and five MA terms. The accuracy of the model was determined using the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). The values of RMSE and MAE for this particular dataset using ARIMA(4,1,5) were 56.70 and 44.44, respectively. The forecast and real-time parameters are shown in Figure 6. Based on the forecast values of the temperature and irradiance and the developed fractional-integral polynomial method for the PV module, the day-ahead PV power for each prosumer (used to trade) in the network is shown in Figure 7.



(a)



(b)

FIGURE 5 Queueing load model results for each agent in the peer-to-peer market. (a) Customers load. (b) Prosumers load.

6.2 | CDA market results

In this section, the CDA market as described in Section 4 was evaluated for the two intervals and four test cases described in the simulation setup. Note that in the graphical results for each case, the clearing mechanism shows the amount of quantity cleared by the DSO. For customers, the clearing mechanism is the demand cleared by the DSO at the retail price that was not traded in the P2P market. Similarly for prosumers, the clearing mechanism shows the excess PV generation bought by the DSO at FIT as that they were not able to sell in the P2P market.

6.2.1 | Case I results: P2P trading without network constraints

Figures 8a and 9a show the results of Interval 1 (low PV generation) for Case I. The excess PV generation of the prosumers is relatively lower compared to the demand of the customers, therefore the prosumers are able to sell a majority of their PV power into the P2P market. On the other hand, the customers are able to meet only a small portion of their demand from the P2P market due to the low PV profile of the prosumers. The remaining demand of the customers is cleared by the DSO at the retail price. Similarly, for the prosumers the remaining portion of the PV generation is cleared by the DSO

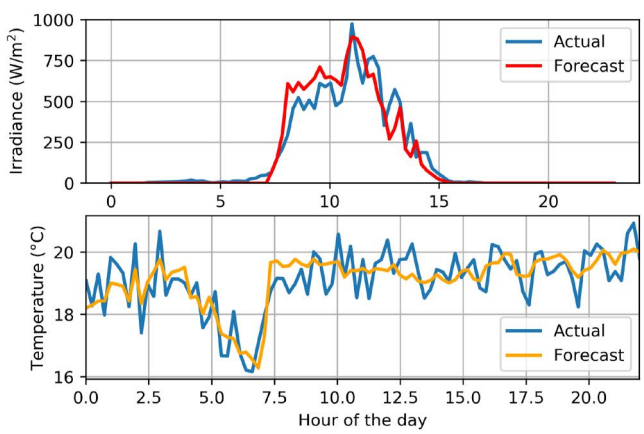


FIGURE 6 Comparison of the forecast results using the Box–Jenkins methodology with the real time parameters. Autoregressive integrated moving average (ARIMA)(4,1,5) was used to obtain the final forecast results.

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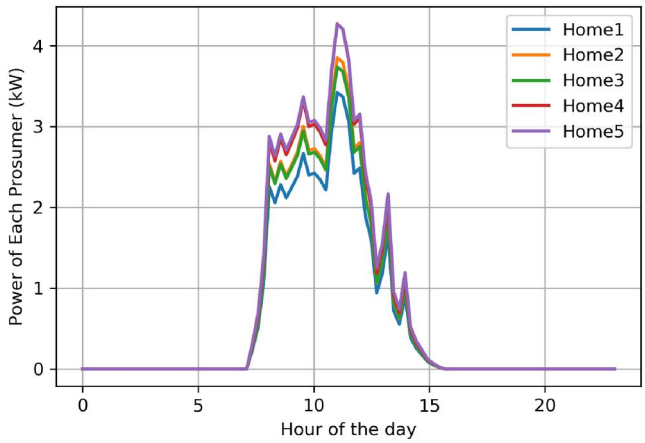
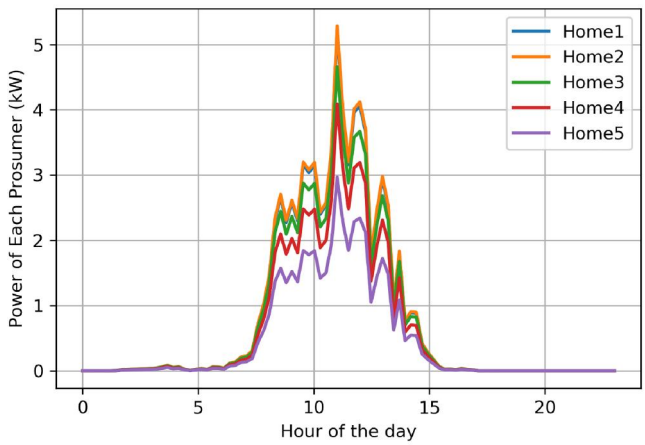
by offering the FIT to the prosumers. Figure 12a shows the welfare metrics for different agents in the market for Case I without losses. Because the majority of the demand of the customers is cleared by the DSO, the customer's welfare is relatively low compared to the prosumer's welfare for Interval 1.

Contrarily, Figures 10a and 11a show the results of Interval 2 (relatively low demand) for Case I. For this particular interval, the demand of the customers is relatively lower compared to the PV generation of the prosumers. Therefore, the majority demand of the customers is traded in the P2P market, and only a small portion of it is cleared by the DSO at the retail price. On the other hand due to the lower demand of the customers and higher excess PV generation, the prosumers are able to sell only a small portion of their PV power through the P2P market while the rest is cleared by the DSO at the FIT. For this particular interval, the customer's welfare is relatively higher as compared to the prosumer's welfare as shown in Figure 12b. All the results are averaged over the total number of the samples to introduce the fairness for each participant of the P2P market.



(a)



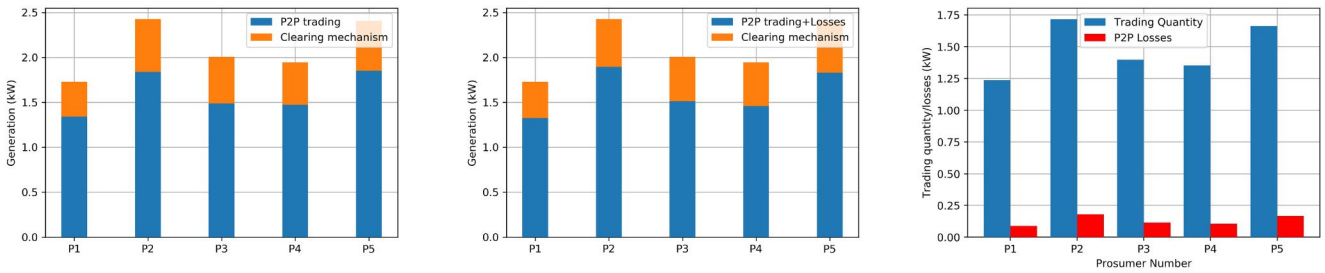
(b)

FIGURE 7 Photovoltaic power of each prosumer in the network. (a) Forecast power. (b) Actual power.

6.2.2 | Case II results: P2P trading with trading losses

Each interval is simulated for Case II while considering the trading losses of the network. Figures 8 and 9b show the results of Interval 1; the same trend can be observed as in Case I, except that now the power balance equation includes both the losses and the traded quantity. A portion of the PV generation of the prosumers is consumed as losses of the network, as shown in Figure 8c. The network losses are directly proportional to the amount of the traded energy between the customers and the prosumers. From the welfare metrics, it is observed that the inclusion of the P2P losses decreases the welfare for each agent of the market, as shown in the Figure 12a. This is due to the fact that a portion of the PV power is now being used to meet the losses of the network.

Similarly, Figures 10b and 11b show the results of Interval 2 for the P2P market while considering the network losses. Again by considering the trading losses, the overall welfare of the P2P participants decreases, as shown in Figure 12b. This highlights a contribution of the work in that P2P markets that



(a) (b) (c)

FIGURE 8 Prosumers energy trading in peer-to-peer (P2P) markets for Interval 1 during different test cases. (a) Trading of the prosumers with customers for Case I and Case III where there are no trading losses. (b) Trading of the prosumers with the customers for Case II and Case IV where the trading losses are included in the network. Note that the amount of energy traded in this case includes both the trading losses and the P2P energy. (c) Breakdown of the losses and traded energy for Case II and Case IV where the trading losses are included in addition to the P2P energy.

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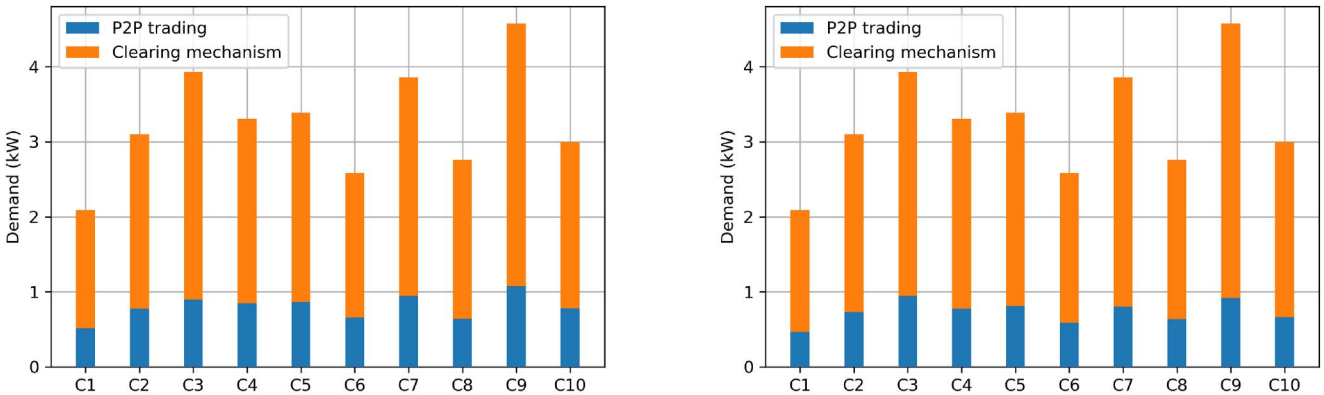
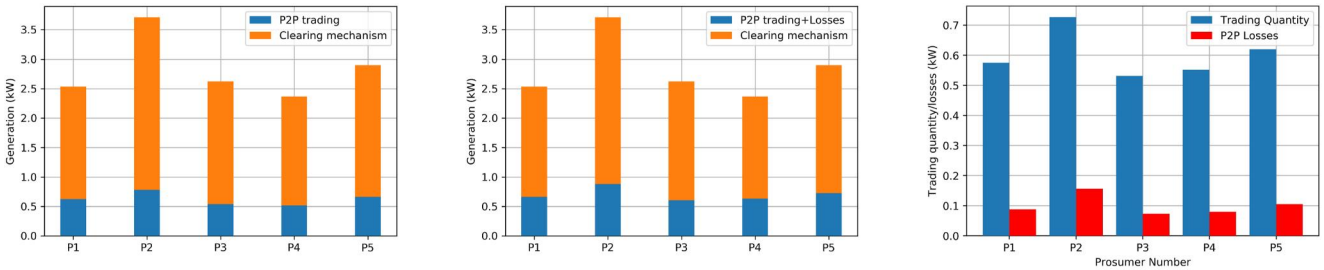


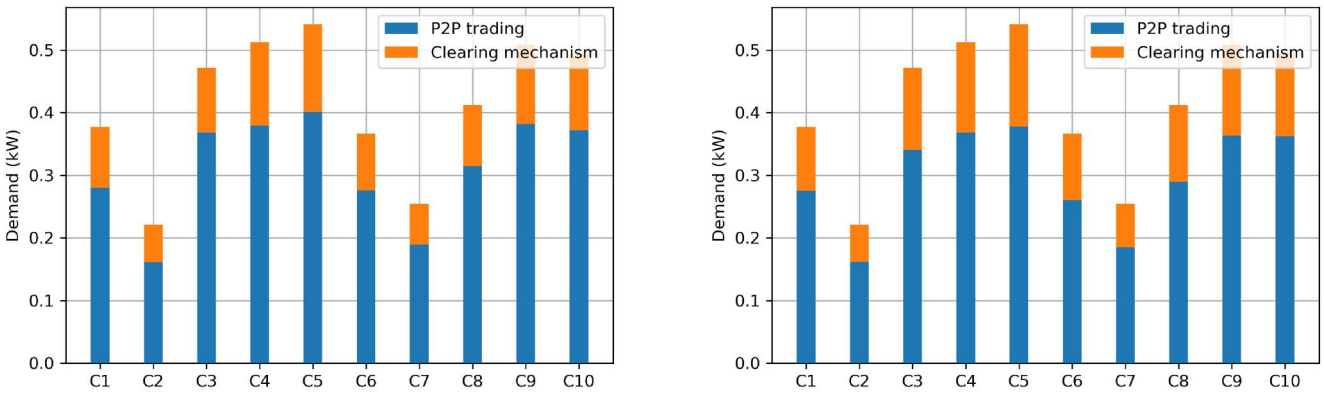
(a) (b)

FIGURE 9 Customers trading with the prosumers for Interval 1 during different test cases. (a) Trading of the customers with prosumers for Case I and Case III where there are no trading losses. (b) Trading of the customers with the prosumers for Case II and Case IV where the trading losses are included in the network. Note that the slight decrease in the amount of the energy traded for Case II and Case IV is due to the losses of the network. The decrease in the amount of traded energy for the customers is in accordance with the losses shown in Figure 8c.



(a) (b) (c)

FIGURE 10 Prosumers energy trading in peer-to-peer (P2P) markets for Interval 2 during different test cases. (a) Trading of the prosumers with customers for Case I and Case III where there are no trading losses. (b) Trading of the prosumers with the customers for Case II and Case IV where the trading losses are included in the network. Note that the amount of energy traded in this case includes both the trading losses and the P2P energy. (c). Breakdown of the losses and traded energy for Case II and Case IV where the trading losses are included in addition to the P2P energy.



(a) (b)

FIGURE 11 Customers trading with the prosumers for Interval 2 during different test cases. (a) Trading of the customers with prosumers for Case I and Case III where there are no trading losses. (b) Trading of the customers with the prosumers for Case II and Case IV where the trading losses are included in the network. Note that the slight decrease in the amount of the energy traded for Case II and Case IV is due to the losses of the network. The decrease in the amount of traded energy for the customers is in accordance with the losses shown in Figure 10c. P2P, peer-to-peer.

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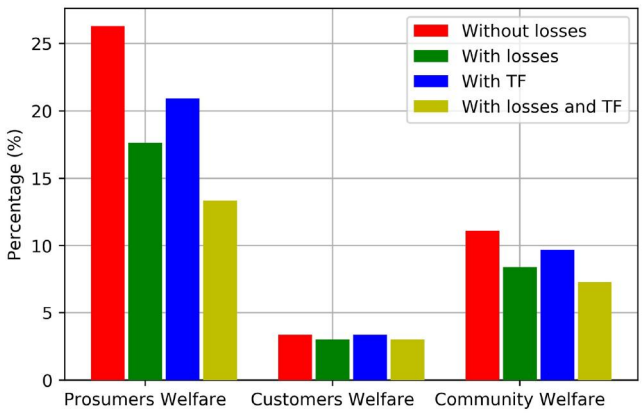
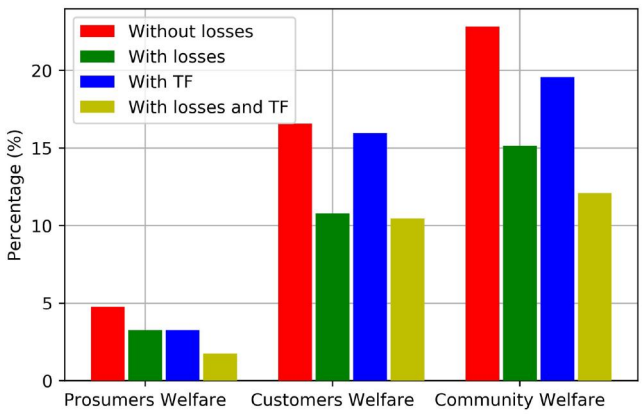
neglect the physical distribution network may overstate the benefits. In the future, this can also lead to more optimal bidding strategies that take into account electrical distance.

6.2.3 | Case III results: P2P trading with the network utilisation fee

Case III for the developed CDA-based market was evaluated including the network fee utilisation model for each prosumer/customer in the network. Because the value of α was kept equal to zero to maximise the incentive of the customers participating in the P2P market, the welfare of the customers is not altered in Case III as evident in Figure 12a,b. Contrarily, the prosumers will be charged for using the network services of the DSO for transferring the energy from one node of the network to the other. Therefore, the welfare of the prosumers decreased for Case III as shown in Figure 12a,b. The remaining trend is the same as the rest of the cases, that is, for Interval 1 the prosumers welfare is relatively higher as compared to the welfare of the customers, whereas the customers benefit more by participating in the P2P market in Interval 2.



(a)



(b)

FIGURE 12 Comparison of the welfare metrics for each case. (a) Interval 1 welfare metrics. (b) Interval 2 welfare metrics. It is evident that the welfare metrics of the agents are effected by incorporating the network constraints. The prosumers welfare is effected by both the trading losses and the network utilisation fee model. Contrarily, due to the value of the contribution factor for the utilisation fee model, the customers welfare is only altered by including the trading losses constraint. A small difference in the customer's welfare for the transaction fee and ideal market is due to the random nature of the bids and ask prices.

6.2.4 | Case IV results: P2P trading with trading losses and network fee

Case IV was conducted by incorporating both the trading losses and network utilisation fee model for the CDA-based P2P market. In this particular case, the welfare of the prosumers was impacted by both the trading and network utilisation constraint. On the other hand, the customer's welfare is impacted only by the trading losses due to the fact that the contribution factor for the utilisation model was still kept at zero. However, the trend for both of the intervals remains the same where Interval 1 benefits the prosumers, and the customers get more welfare during Interval 2. Figure 12a,b show the comparison of the welfare metrics for each interval for Case IV. The prosumers welfare for Case IV is the lowest due to the fact that both the trading and network fee utilisation constraints are added to the system. However, the customer's welfare is not altered due to the network utilisation constraint because of the value of α_c.

6.3 | Overall simulation results

The customers, prosumers, and the community welfare metrics are evaluated for each case during the solar hours (i.e. when there was solar irradiance and, hence, the market could be conducted). Figure 13 shows the overall results of the welfare metrics during the P2P trading hours. Figure 13a shows the welfare of the customers for each developed test case during the P2P trading hours. For the intervals where the excess PV generation of the prosumers given in the form of bar graphs is lower, the welfare of the customers will decrease. This is due to the fact that during such intervals the customers will only be able to meet a small portion of their demand from the P2P market while the remaining would be cleared by the DSO at the retail price. Contrarily, for the intervals where the PV generation is relatively lower, the prosumers are able to sell majority of their power via P2P trading and only a small portion would be cleared by the DSO at the FIT. This gives relatively higher prosumers welfare for such intervals as evident from the Figure 13b. The community welfare is sensitive to both the prosumers and customers welfare as shown in the Figure 13c, however in all cases the implemented market was beneficial to the community. Figure 14 shows the comparison of the retail and FIT with the average P2P price for each case during the solar hours. It is evident from the figure that the suggested energy trading platform offers a competitive price for the

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market participants compared to the retail and FIT. Prosumers are provided a local price that is higher than the FIT, whereas the customers get a lower price compared to the retail tariff. This results in a positive welfare for both the customers and prosumers participating in the P2P energy trading platform.

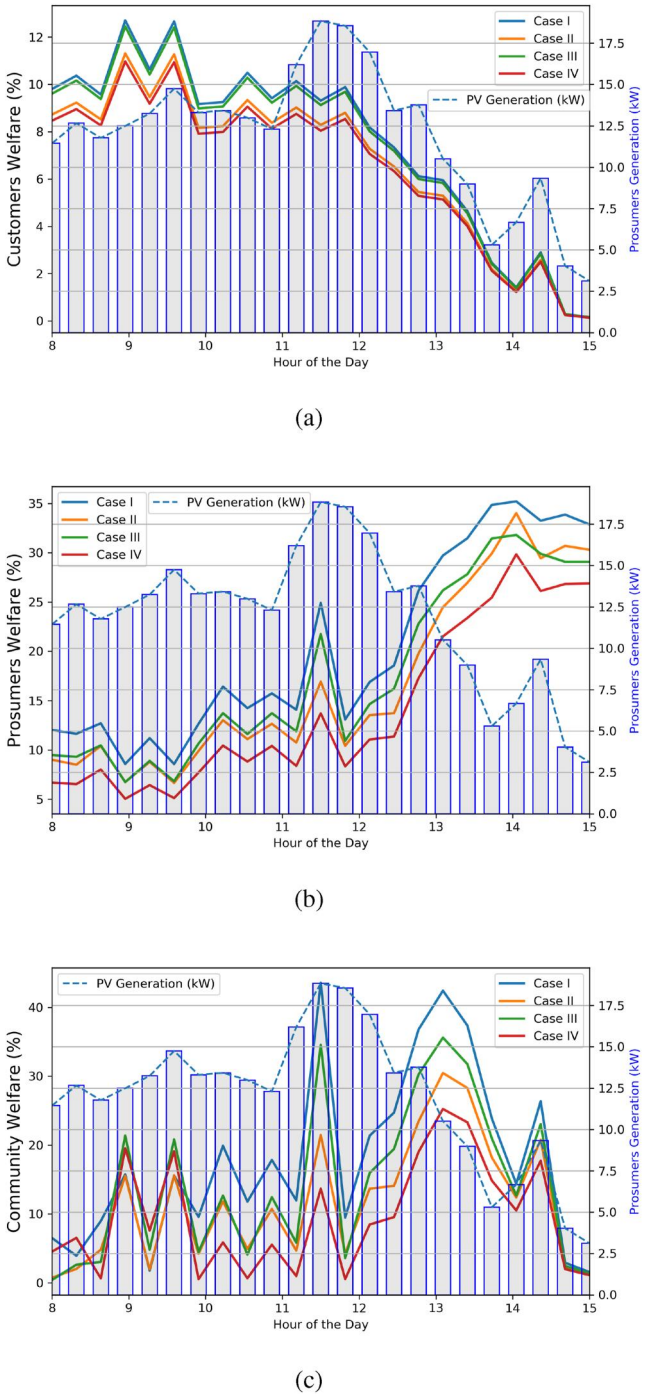

FIGURE 13 Welfare metrics for each agent in the market for different test cases. (a) Customer's welfare. (b) Prosumer's welfare. (c) Community welfare. The results are shown as the double axis graph where the bar graph represents the total amount of the prosumers excess photovoltaic (PV) generation with respect to the generation axis.

6.4 | Comparison with real-time parameters and savings of market participants

The last step in the analysis of the suggested CDA-based market is to compare the performance of the day-ahead forecast results with the real-time parameters [49, 50]. As the forecast was used for the day-ahead P2P market design, we compared the error in our forecast to that of perfect information. This difference can be thought of as the cost of the forecasting error (i.e. how much benefit would be obtained via a perfect forecasting method). This research uses the mean comparison of the traded energy and welfare metrics computed over all the P2P intervals. Table 1 shows the results of the mean comparison for forecast and real-time parameters. From the table, it is evident that the forecast and real-time markets are close which justifies the use of the day-ahead forecasting algorithms and PV modelling to develop the day-ahead CDA-based P2P market. The small mismatch between the forecasted and actual markets can be incorporated by introducing intra-day trading for the given market structure. However, this aspect of the suggested constrained CDA-based market has been left for future work.

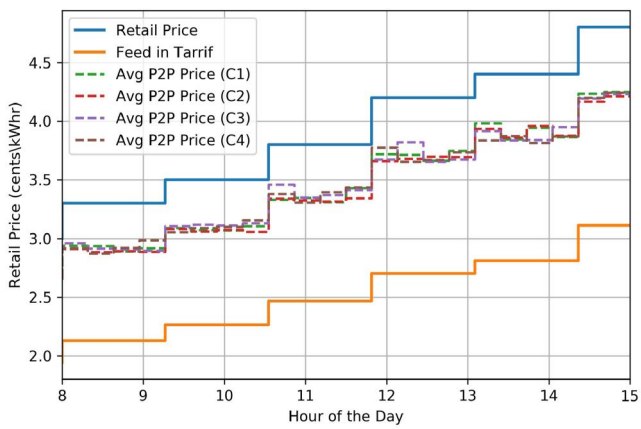

FIGURE 14 Comparison of retail and feed-in tariffs (FIT) with the average peer-to-peer (P2P) price for each case during the solar hours.

TABLE 1 Mean comparison comparing the forecast and actual results for each case.

Parameters	Case I	Case II	Case III	Case IV
Mean traded energy comparison (kW)				
Forecast	3.878	3.489	3.716	3.537
Actual	3.715	3.385	3.826	3.801
Mean customers welfare comparison (%)				
Forecast	7.5154	6.688	7.365	6.489
Actual	5.8530	5.583	6.104	5.864
Mean prosumers welfare (%)				
Forecast	20.375	16.854	17.604	14.141
Actual	20.524	16.534	17.784	14.359

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TABLE 2 Monthly savings of the market participants.

Market participant	Case I	Case II	Case III	Case IV
Savings of the market participants ($)				
Customers savings	38.00	35.87	37.87	34.67
Prosumer savings	42.50	33.91	36.91	28.91

To find the economic benefits of the suggested constrained CDA-based P2P market, the developed model was used to find the savings of the market participants for the entire month of July, 2016. The original monthly bill for the customers without participating in the market is $186. The results of the savings for each case are summarised in Table 2. On average, a customer saved around $3.80 and a prosumer $8.50. While the monetary benefits may seem to be small for the suggested market, it is extra income which the participants can earn without having to impair their comfort level (i.e. no loads have been impacted). Similarly, if we scale the suggested market to a larger test system and consider different geographic locations where there is higher retail tariff, the suggested market will result in more revenue for the market participants.

7 | CONCLUSIONS

This paper presented a day-ahead model for a CDA-based P2P market framework while incorporating network constraints. The queueing load model was used to develop the demand curves for all market participants. Additionally, the combination of an ARIMA model and fractional and integral polynomial methods were used to determine the forecast PV power for each prosumer in the network. The CDA market was evaluated while considering the trading losses and network fee utilisation model to capture the physical constraints of the network. It was observed that the physical constraints of the network decrease the overall welfare of the P2P agents. Additionally, the results of the day-ahead market were compared with the real-time parameters. The difference was small, and any such differences could be rectified in intra-day trading. The future work includes extending the current model to include the voltage constraints while developing the OPF equations for the network and using a Game Theoretic approach to develop the bid/ask values for the CDA market and other distributed energy resources to the suggested framework.

AUTHOR CONTRIBUTIONS

Sheroze Liaquat: Conceptualisation; Data curation; Methodology; Software; Visualisation; Writing – original draft. **Tanveer Hussain**: Data curation; Methodology; Validation; Writing – review & editing. **Berk Celik**: Data curation; Validation; Writing – review & editing. **Robert Fourney**: Data curation; Supervision; Validation; Writing – review & editing. **Timothy M. Hansen**: Conceptualisation; Data curation; Methodology; Supervision; Validation; Writing – review & editing.

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NOMENCLATURE

$B_{ask,\Delta t}$	ask values for prosumers
$B_{bid,\Delta t}$	bid values for customers
c	number of customers
C_c, C_p	original market prices
C'_c, C'_p	market prices with P2P trading
$C_l(t)$	distribution load data
c_p	trading price
d	duration of appliance
d_{ij}	electrical distance
$D_{\Delta t}$	demand of participants
E_{cl}	clearing quantity
E_T	total traded quantity
G_{stc}	irradiance at standard test conditions (STC)
$G_{\Delta t}$	surplus PV generation of prosumer
i	customer index
i_{opt}	optimal current value
i'_{sc}	short circuit current
I_t	short circuit temperature coefficient
j	prosumer index
k	time series index
$L(t)$	aggregated load of individual customer
$l^k(x_t)$	lag operator
p	power of appliance
p_{ij}	P2P quantity
P_L	network losses
p_m	parallel connected modules
p_r	number of prosumers
s_m	series connected modules
$TF(p_{ij})$	total network utilisation fee
$T_{feed-in,\Delta t}$	feed in tariff
$T_{retail,\Delta t}$	retail price
T_{stc}	temperature at STC
$v_{\max}$	maximum voltage of module
$v_{\min}$	minimum voltage of module
v'_{oc}	open circuit voltage
V_t	open circuit temperature coefficient
x_t	time series data
z_{ii}, z_{jj}	self impedance
z_{ij}	mutual impedance
$Z_{ij,equ}$	equivalent impedance
α_c	contribution factor
$\beta_{\max}$	maximum household load
$\beta_{\min}$	minimum household load
γ_i, γ_j	charge rate coefficients

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Γ_{ij}	loss coefficient
δ	parameters of MA model
Δt	market interval
ϵ	white noise error terms
η_c	customers welfare
η_{comm}	community welfare
η_p	prosumers welfare
$\lambda(t)$	appliance rate
τ_t	trading matrix
ϕ	parameters of AR model

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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